

PSY 3307 Slides

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Week 1 - Introduction & Review

Agenda

- ▶ Introductions
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- ▶ Review
- ▶ Types of Research
- ▶ Steps of Conducting Statistics Within the Research Process
- ▶ Levels of Measurement
 - ▶ Measurement Error
- ▶ Measurement Scales

About Me

- ▶ My name is Jonathan A. Pedroza (he/him)
 - ▶ please refer to me as **JP** or Jon
- ▶ First-generation Graduate
 - ▶ CPP Alumni (2014)
 - ▶ CSUDH (2017)
 - ▶ University of Oregon (2021)

About Me



About Me



About Me

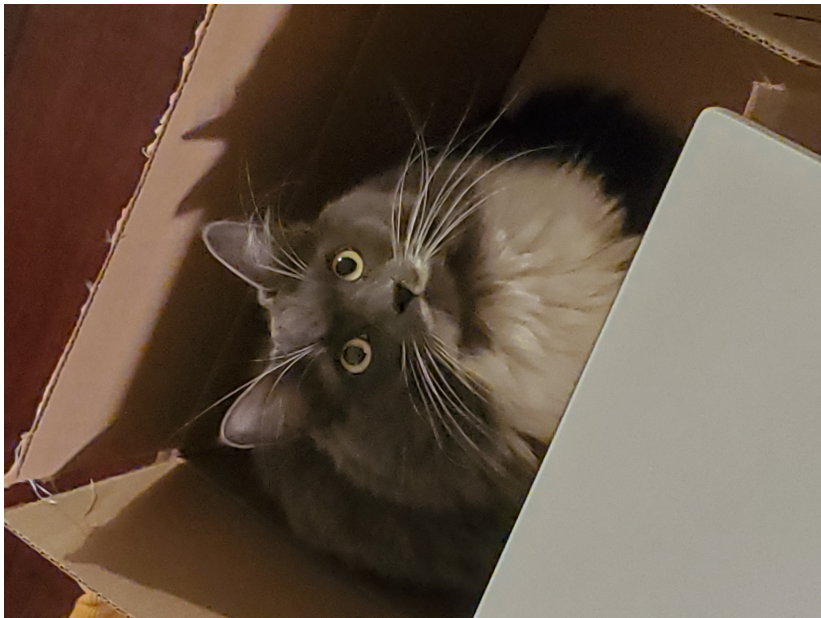


Figure 3: lucy

Nerdy Stuff

- ▶ Passion/Nerdy Topics:
 - ▶ I love creating data visualizations in R
 - ▶ health inequities in communities of color
 - ▶ examining environmental variables that influence health behaviors

Nerdy Stuff

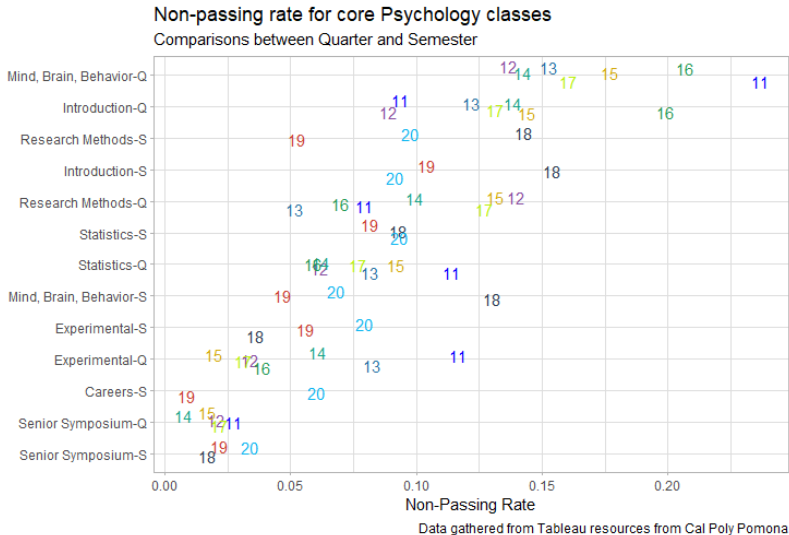


Figure 4: non_passing_rate

Break-out Rooms - Tell Me About Yourself

- ▶ Name & pronouns
- ▶ What do you want to do after you graduate?
- ▶ What is your experience with statistics?
- ▶ What is your experience with SPSS?
- ▶ Something that you are passionate/nerdy about

Learning Objectives

- ▶ Use SPSS to compute statistics
- ▶ Read/Interpret Descriptive & Inferential Statistics
- ▶ Understand & Solve Formulas for Statistical Tests
- ▶ Understand Sampling, Statistical Null Hypothesis Testing, Effect Sizes, Confidence Intervals
- ▶ Demonstrate ability to apply these skills to test hypotheses using real data

Topics Covered

- ▶ Descriptive Statistics
- ▶ Probability
- ▶ Random Selection
- ▶ Normal Curve & Sampling Distribution
- ▶ Hypothesis Testing
- ▶ Rules for Statistical Decisions (Types I and II Errors)
- ▶ Sample Size Importance (One- vs Two-tailed Tests)
- ▶ Z-scores, z-ratio, and t-ratio (single group, independent-samples t-test, paired)
- ▶ Analysis of Variance (One-way, Two-way, Repeated-measure, mixed)
- ▶ Confidence Intervals/Error bars in plots
- ▶ Most common plots (histograms, boxplots, scatterplots)
- ▶ Correlation & Regression Analysis
- ▶ Chi-square (One-way, Two-way)
- ▶ Non-parametric statistics